

Manuel Alejandro Mosquera Ortega

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Profile

PhD student in Machine Learning and Artificial Intelligence, researching methods to enhance the reasoning and planning capabilities of AI agents. My work focuses on the intersection of world models and energy-based models for planning and decision-making. I have strong expertise in large language models, reinforcement learning, and multi-agent systems. I am the first author of peer-reviewed research published at top-tier venues (AAAI, IEEE Transactions on Artificial Intelligence) on LLM-based autonomous agents and low-resource machine translation. I have demonstrated experience applying reinforcement learning techniques, including policy optimization methods such as GRPO, and designing agent-based architectures evaluated in simulation environments.

Research Experience

Improving Low-Resource Translation with Dictionary-Guided Fine-Tuning and RL: A Spanish-to-Wayunaiki Study, 2025: First author of a paper accepted at AAAI 2026. This paper demonstrates how recent reinforcement learning techniques like GRPO can help teach an agent to use a bilingual dictionary, improving translation performance. Available on: <https://doi.org/10.1609/aaai.v40i46.41251>

Can LLM-Augmented Autonomous Agents Cooperate? An Evaluation of Their Cooperative Capabilities Through Melting Pot, 2025: First author of the paper. This work explores how state-of-the-art language model-based agent architectures perform in multi-agent environments and how they compare to reinforcement learning agents available in Melting Pot. Available on IEEE Transactions on Artificial Intelligence: <https://doi.ieeecomputersociety.org/10.1109/TAI.2025.3569192>

Master's Thesis: Cooperative Agents with Large Language Models (LLMs). The recent advancements in large language model capabilities have amazed the world with their usefulness in solving multiple tasks, often matching or even surpassing human performance in many areas. These models have enabled the creation of agents—autonomous systems that utilize language models as their core intelligence. Along with memory structures, planning modules, and action mechanisms, these agents can navigate different environments to not only respond to requests but also complete tasks and achieve objectives. As these agents are deployed in real-world applications, studying cooperation in multi-agent environments with language model-based agents becomes crucial for understanding and improving the outcomes of their interactions.

Education

2025 –	PhD in Engineering, Universidad de los Andes. Researching reasoning in AI models.
2022 – 2025	Master's in Systems and Computing Engineering, Universidad de los Andes with an emphasis on artificial intelligence. Cumulative GPA: 4.85/5.00
2018 – 2019	Specialist in Financial Management, Universidad de los Andes. Cumulative GPA: 4.72/5.00
2012 – 2017	Business Administrator, Universidad del Rosario. Cumulative GPA: 4.52/5.00

Professional Experience

2025 –	Professor, Universidad de los Andes. I teach the Introduction to Programming course in the Faculty of Engineering.
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Professional Experience (continued)

- 2023 – 2024 **Teaching Assistant**, Universidad de los Andes. I developed various tutorials, graded exercises, and workshops for the courses “Fundamentals of Natural Language Processing” and “Advanced Models of Natural Language Processing” as part of the master’s program in Artificial Intelligence at Universidad de los Andes.
- 2021 – 2023 **Frontend Web Developer**, Rappi. I developed user interfaces for two Rappi web portals using JavaScript, TypeScript, and React to produce high-quality code and scalable solutions.
- 2015 – 2021 **Entrepreneur**, Cartongraficas SAS. A family business in the graphic arts sector where I contributed mainly to the company’s top management, as well as optimizing processes through the development of applications in Excel and VBA.

Achievements and Other Studies

Achievements and Recognitions

- 2025 **Graduated cum laude** for achieving a weighted GPA within the top 3% of the last 5 years in the Master’s Program in Systems Engineering.
- 2019 **Best GPA** of the nineteenth cohort of the Specialization in Financial Administration, Universidad de los Andes.
- 2017 **Ranked 9th out of 98** graduates of the Business Administration program, class of 2017, Universidad del Rosario.

Certifications

- 2023 **AI for Medical Diagnosis**. DeepLearning.AI.
- 2022 **Building Transformer-Based Natural Language Processing Applications**. NVIDIA DLI.
Fundamentals of Deep Learning. NVIDIA DLI.
Mathematics for Machine Learning, 3 courses. Imperial College London.
Accelerated Computer Science Fundamentals, 3 courses. University of Illinois at Urbana-Champaign.
- 2021 **The Arduino Platform and C Programming**. University of California, Irvine.
Introduction to the Internet of Things and Embedded Systems. University of California, Irvine.
Introduction to Calculus. The University of Sydney.
Intro to Machine Learning. Kaggle.
- 2020 **Crash Course on Python**. Google.
- 2017 **Introduction to Data Science in Python**. University of Michigan.
Python for Everybody, 5 courses. University of Michigan.

Skills

Languages	Advanced oral, reading comprehension, and writing skills in both English and Spanish.
Programming Languages	Python, JavaScript, NodeJS, R, and C++.
Libraries and Frameworks	Pytorch, Keras, Pandas, NumPy, React.